drawing the line

digitalisierung und programmierung · prof. dr. nicolas meseth

1 a world without steps

2 two cuts and a price

3 drawing your own lines

part 1

a world without steps

is this digital?

image: ai-generated (gpt-image-2)

digital is not the same as electronic

digital

flip scoreboard

light switch

abacus

a die

analog

slide rule

dial thermometer

dimmer

hourglass

not one of them contains a chip.

images: ai-generated (gpt-image-2)

digital means finitely many states, and nothing in between counts.

both halves carry weight. the second one is where the work is.

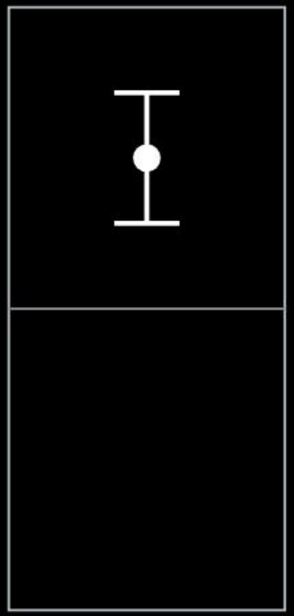
the part no program controls

light is analog:
between any two brightnesses lies another.

light is analog:
between any two brightnesses lies another.

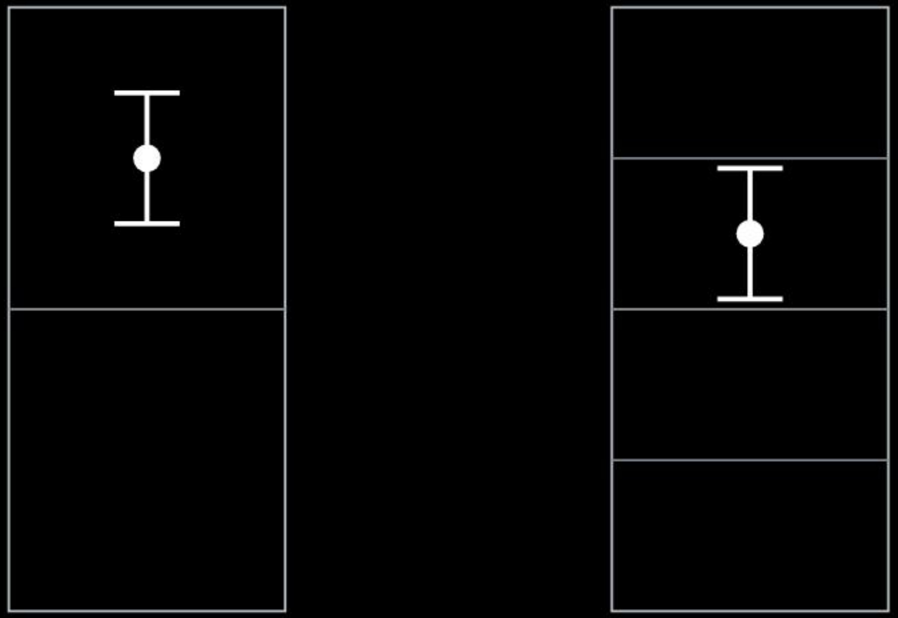
your program needs states:
finitely many, clearly apart.

same noise, more regions



2 regions

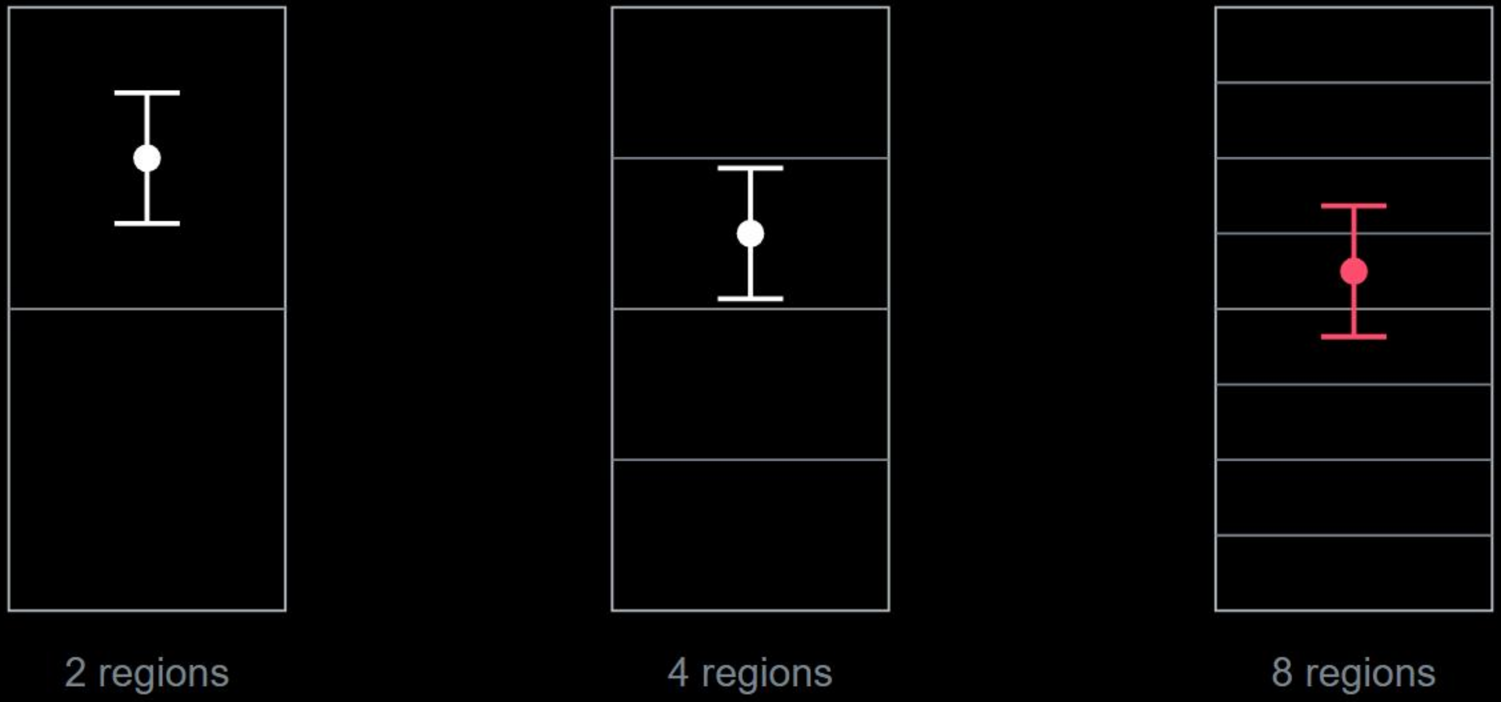
same noise, more regions



2 regions

4 regions

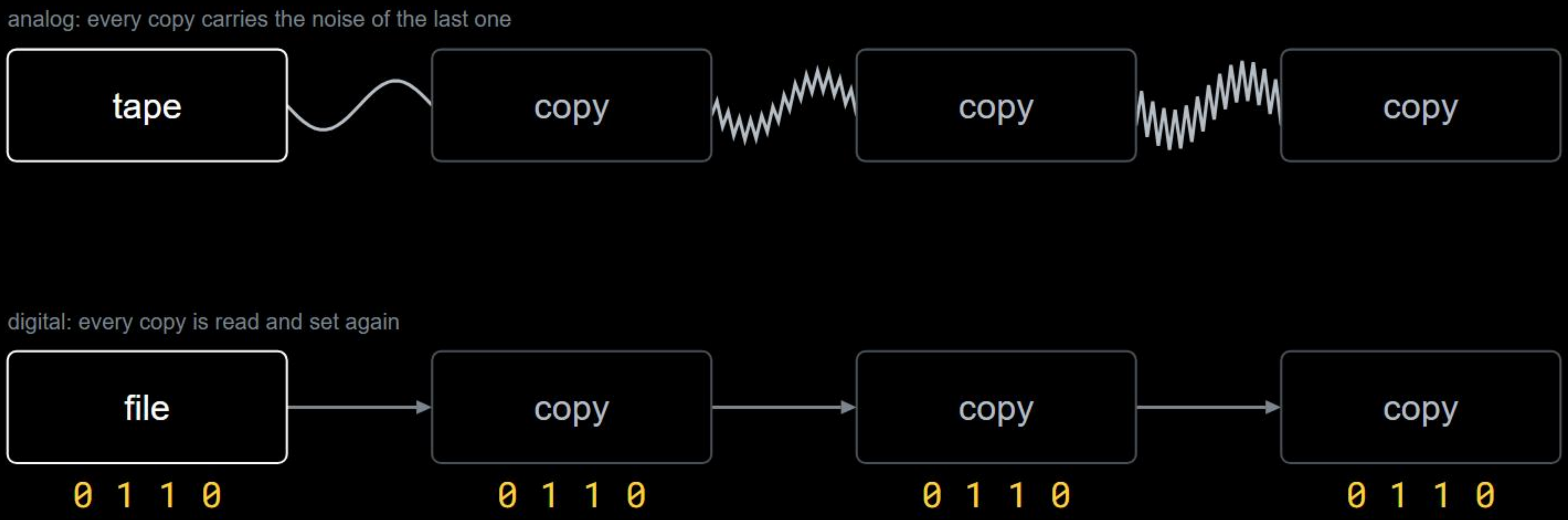
the same measurement with the same wobble, three times same noise, more regions



the same measurement with the same wobble, three times

the noise did not grow. the margin shrank.

what the thrown-away precision buys



every digital copy is a rebirth from clean states.

part 2

two cuts and a price

digitising throws information away.
on purpose.

which exact value it was inside the region: gone for good.

digitising takes two cuts

the analog picture

16 × 16 points

cut 1: sampling

how many points? that is the resolution

digitising takes two cuts

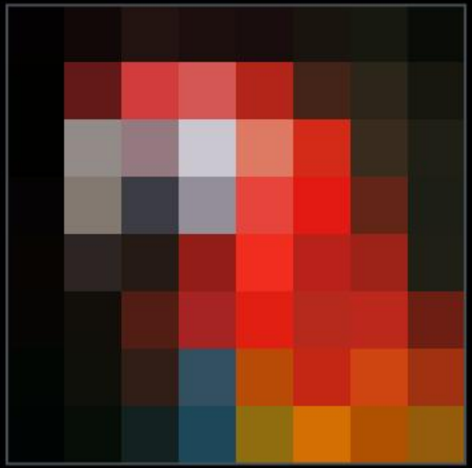
the analog picture | 16 × 16 points | 4 grey levels

cut 1: sampling
how many points? that is the resolution

cut 2: quantization
how many levels? that is the colour depth

sampling slices the axis. quantization slices the value.

which cut was too coarse?



too few points

blocks

too few levels

hard patches instead of soft shading

`64 × 64 points × bits per point = size`

points	bits per point	size	over the light link at 30 bit/s
4,096	24 bit	12 KB	about 55 minutes
4,096	1 bit	512 bytes	about 2 minutes

what you keep, you pay for in minutes.

how far can you go?

128 × 128
49,152 bytes

how far can you go?

128 × 128
49,152 bytes

32 × 32
3,072 bytes

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128 × 128
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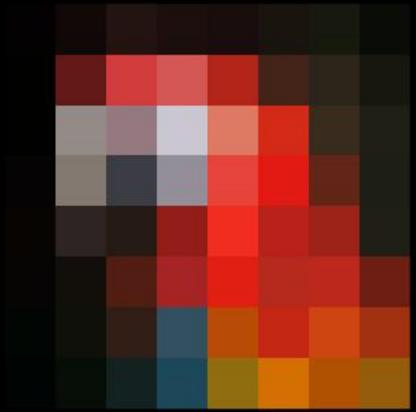
32 × 32
3,072 bytes

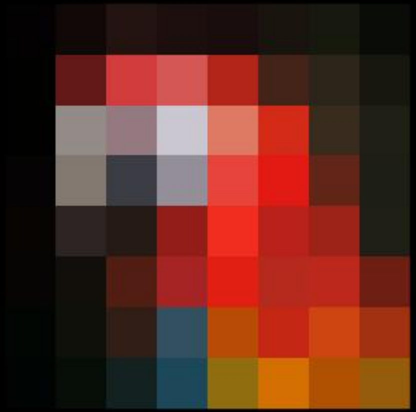


8 × 8
192 bytes

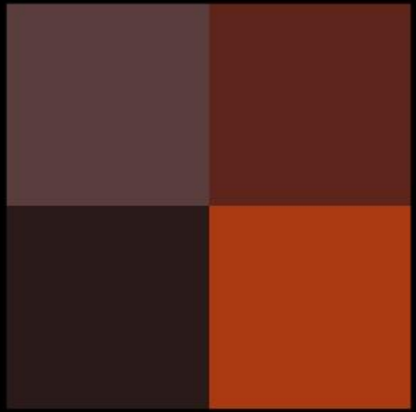
how far can you go?

128 × 128
49,152 bytes

32 × 32
3,072 bytes



8 × 8
192 bytes



2 × 2
12 bytes

where would you stop?

try it: what would you throw away?

picture:



the analog picture (endless detail)

your digital picture

16 × 16 pixels × 2 bit

= 512 bit = 64 bytes

over your light link at 30 bit/s: about 17 seconds

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resolution (how many pixels)

16 × 16 = 256 pixels

colour depth (how many colours per pixel)

b/w | 4 greys | 4-bit | greyscale | 8-bit | 16-bit | 24-bit the same two cuts, for sound

sampling slices time.
quantization slices loudness.

44,100 samples per second at 16 bit: that is a cd.

part 3

drawing your own lines

```
value = sensor.get_clear()

if value > 250:
    symbol = "bright"
else:
    symbol = "dark"
```

the sensor delivers hundreds of numbers. this line turns them into two symbols.

how many colours are safe?

two are safe. eight are fast.

how many colours are safe?

two are safe. eight are fast.

somewhere in between sits your alphabet, and only your own measurements know where.

you decide where the lines go.

you decide where the lines go.

from there on, your decision is the truth, not the physics.